

A SMALL MAGAZINE ABOUT A VERY LIVE LANGUAGE

Week in

HAZEL

ISSUE 001

04-10 SEPTEMBER 2026

PILOT / ABOUT 12 MINUTES



The cabinet is open.

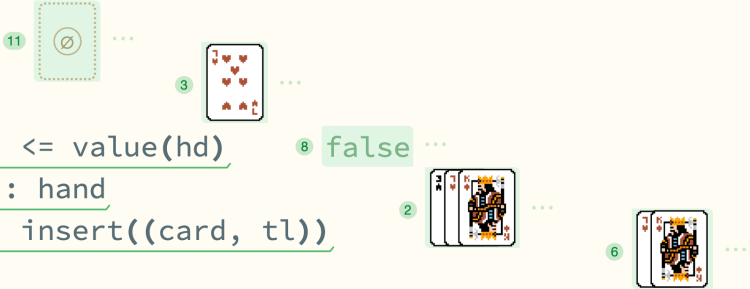
Probes IV lands. Modules grow another floor. A foreign runtime moves in. The experiments have experiments.

MERGED INTO DEV / PROBES IV

A little more room to see what happens.

Insertion Sort

```
let insert(card : Card, hand : Hand) : Hand =
  case hand
  | [] => [card]
  | hd :: tl =>
    case value(card) <= value(hd)
    | true => card :: hand
    | false => hd :: insert((card, tl))
  end end in
```



```
let sort(hand : Hand) : Hand =
  case hand
  | [] => []
  | hd :: tl => insert(hd, sort(tl))
  end in
```



Lossless **dev** capture. Insertion sort, with intermediate hands displayed as cards. The renderer arrived here from Alexander's work; these are runtime values beside the program.

On September 8, Andrew Blinn's long-running **Probes IV** branch became part of **dev**, merged by Cyrus Omar. The headline is a drawer: a sample can unfold into a multiline view below its source line. A cramped chip gets a room of its own. Each drawer remembers its open state and scrolls after fifteen rows, keeping a large value attached to the line that produced it.

Auto-probe also grows from a switch into **Off / Caret / All**. Rich views can appear inside the sample display, and a quieter green palette makes focus easier to follow.

The less photogenic fixes matter just as much. Newly placed probes should show values without waiting for another edit. Step-into reaches indirect calls. Sample navigation scrolls with the selection. Menus move from overeager hover to right- or alt-click.

Caret mode follows one top-level definition; All probes the program, one probe per source row. Choose between inspecting one neighborhood and surveying the whole street. Try **Documentation** → **Cards** on dev. Then open a sample drawer. This is the week's biggest change to the everyday feel of Hazel.

[#2363: features, review & merge](#) / [Try dev](#)

FOLLOW-UPS / STILL OPEN

And then the drawer had a few complaints.

A landing is an event. Settling in takes longer.

01 / INTERACTION

Menus lose their footing.

Reports #2519-2521 describe rich-table drawer trouble: a hidden table still has a “Hide table” menu, sample context menus are hard to reach, and a column menu can anchor below the whole drawer. Small slips in where controls appear become large interruptions when you are following a value.

02 / PERFORMANCE

Recursion gets expensive.

Issue #2524 reports a Probes IV regression: ascription bookkeeping grows with call-stack depth. In the reporter’s CLI example, 12,800 recursive calls run out of memory after the merge; the pre-merge version completes. This is a reported regression, not an independent benchmark.

03 / TEST SIGNAL

A green build with an asterisk.

Merged #2482 disables two intermittent evaluator properties, keeps their bodies compiling, and adds deterministic checks for known disagreements. The underlying bugs remain open. The test signal changed; those bugs did not vanish.

AT THE WORKBENCH

Matthew’s review of #2482 says a good Fumola change had been reverted after an unrelated flaky test failed. That gives a noisy test a very concrete cost: a useful change is removed, someone must trace the failure, and the experiment waits. Better diagnostics protect branch work as well as dev.

12

issues opened

4

issues closed

344

issues outstanding*

Flows: September 4–10 UTC. *Open backlog at the September 11 revision capture; not a historical end-of-week reconstruction.

[#2519](#) / [#2520](#) / [#2521](#) / [#2524](#) / [#2482 & review](#) / [Open issues](#)

NINE DRAFT PRS / NOT A RELEASE

Modules, all the way up.

Alexander Bandukwala has built a nine-part staircase from real module values to a modular standard library. The higher floors are explicitly provisional.

A READING MAP, NOT NINE INDEPENDENT FEATURES

1-3

A module gets an identity

Signatures, width matching, abstract members

#2493 #2494 #2495

4-7

Make the edges behave

Module functions, errors, expectations, duplicates

#2498 #2500 #2501 #2503

8-9

Try the language on itself

Modular implicits + a modular standard library

#2508 #2510

The useful shift: a module becomes a value with a signature, rather than a dressed-up labeled tuple. Width matching lets a richer module satisfy a smaller interface where an expected type is given. Abstract members give names such as **M.T** a meaningful boundary.

For example, a greeting function that asks only for **name : String** can accept a module containing both a name and an age. The narrower interface hides the extra member. Callers can add information without rebuilding a tiny module just to fit the function.

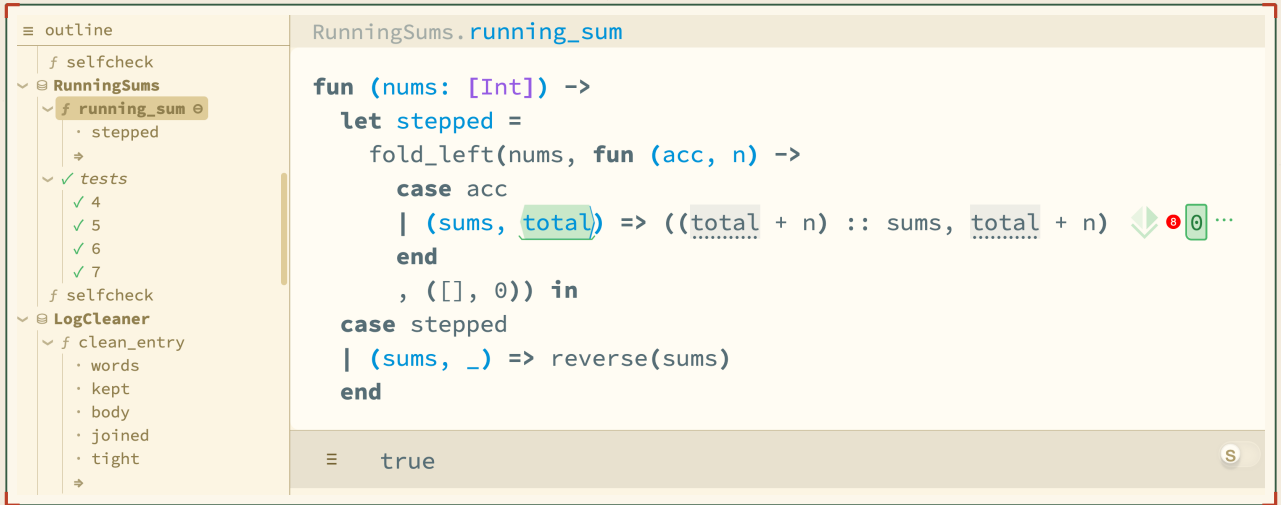
[Start at #2493](#) / [Read the caveats in #2508](#) / [Library sketch #2510](#)

That generosity has a boundary. Differently shaped modules in the two arms of an **if** do not silently acquire a common interface; the author must write the narrowing ascription. The place where information disappears stays visible in the source.

Upstairs, implicit module arguments and **List.map**-style names ask how the language feels using its own machinery. Parts 8 and 9 say they are not close to done. Sharing constraints, implicit-design questions and unmeasured library costs remain part of the story.

DRAFT: MODULAR EDITORS / OPEN: AGENT HARDENING

Give the work a place to happen.



Lossless capture, staged in **modular-editors**. One function from Mega 1k, eight probe samples, and the whole-program result **true**. We added the probe. This is an experimental branch.

Andrew's **modular editors** let a definition occupy the editing surface while its program supplies context, results and tests. A recent fix makes edits beyond the first definition feed whole-program evaluation.

The screenshot is a small desk inside a large room: **RunningSums.running_sum**, lifted out of Mega 1k. Its accumulator builds partial sums in reverse order, then reverses the list. The probe exposes an intermediate total; the result below still belongs to the surrounding program.

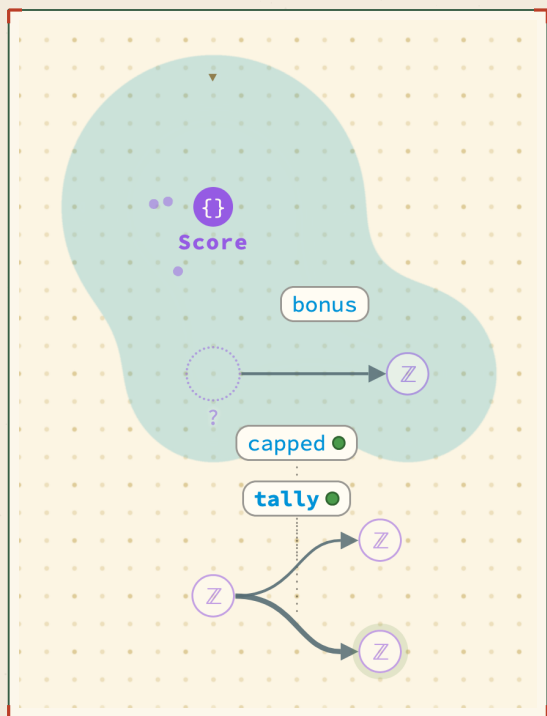
That distinction matters for an assistant too. Selecting a smaller piece should not mean silently changing the world in which it runs. The open **agent-hardening** PR makes probe requests wait for evaluation to settle, bounded at roughly ten seconds. Otherwise an assistant may inspect a value before its latest work has finished.

The canvas stack brings these ideas together: a graph to find the work, a definition-sized editor to do it, and observations that still refer to the whole document. The next page shows the map running locally.

A useful test of the idea: pick a definition, inspect a live sample, then return to the whole program. The interfaces must agree about which document the value came from.

DRAFT #2505 / GRAPH, EDITOR, AGENT

The map has somewhere to go.



Native PNG from the local build. **Scorekeeper Module:** a hull holds **Score** and its member **bonus**; **tally** and **capped** have passing-test dots. Tight crop; no synthetic nodes.

Constellation is becoming an editing surface. Types are nodes; functions run between them. In this demo, **Score** supplies a base of 10, a doubling bonus and a cap of 50. The little program returns 24 for an input of 7. The graph gives those definitions a geography.

This week's new **canvas mode** puts the map in the main area and the selected definition in a panel below. We opened **tally** and the Counter App's **update** this way. The panel also offers observed values and tests. Leaving the mode restores the whole program.

A subsequent fix makes sample interactions act on the master editor, and outlines the selected sample. A value well should stay attached to the running document even while a definition has its own desk.

THE AGENT NEEDS STAGE DIRECTIONS

The agent-facing work gives edits a visible sequence. **Pace** separates bursts into travel, act and settle; **follow** frames the work site. This week's commits make module hulls glide with their nodes and let the camera hold when the program is already in view. The aim is a change you can follow without chasing it.

The motion claims come from code and toolbar; we did not run an LLM session. We inspected the graph and definition panel locally.

There is plenty to try: six demos include nested modules, many callers and pipeline depth. The PR is still marked draft, not ready for review, and its stacked diff includes several other branches.

[Canvas #2505](#) / [Canvas mode](#) / [Sample fix](#) / [Hull motion](#) / [Camera hold](#)

OPEN PR / FUMOLA LIVELITS / EXPERIMENTAL

Two languages. One notebook.

TWO LANGUAGES, ONE NOTEBOOK

```
let gcd : Int =  
  ^fumola_put_force(0, "`zine_gcd", "Gcd.gcd(12, 18)") in  
  
let answer : Int =  
  ^fumola_eval(0, "`zine_counter := 41; get(`zine_counter) + 1") in  
  
(gcd, answer)
```

≡ (6, 42)

A staged, running example on **fumola-livelit-mvp**: Fumola calculates a GCD and reads a named cell; Hazel receives **(6, 42)**. The shot shows the bridge working. It does not measure incremental reuse.

Matthew Hammer's Fumola integration puts a foreign runtime inside a Hazel document. The interesting boundary is ownership: the livelit carries a name for state that lives elsewhere. Sharing an instance can mean sharing its store and history. Copying or reloading the document raises questions an ordinary slider never had to answer.

The branch has already outgrown its original two-widget PR description: its current documentation names four livelits. The interface is still moving.

Two loader PRs merged this week, both **into the Fumola feature branch**. They add source fallback and content-addressed loading, plus a way to identify the runtime that actually loaded.

Small infrastructure, large consequence. The runtime lives outside Hazel's repo and can change independently. The branch's runtime notes explicitly identify gaps in live integration testing. A content hash makes a surprise version diagnosable; it does not make the dependency pinned.

[#2492](#) / [Branch merges #2504](#) / [#2509 & runtime notes](#) / [Try the preview](#)

RECURRING CHARACTERS / THIS WEEK'S THREADS

Around the workbench.

Andrew Blinn

@disconcision

Probes IV reaches dev; modular editors, agent hardening and the canvas keep stretching the workspace around it. The recurring question: how much more of a live program can we keep in view?

Alexander Bandukwala

@7h3kk1d

Modules II is the big construction site. Alongside it: tutorial authoring, precise error reports, CI hygiene and the first reports of several probe regressions. A language grows; its seams become visible.

Matthew Hammer

@matthewhammer

Fumola livelits and the Wasm spike both cross runtime boundaries. Loader fixes, benchmark corrections and a review tracing a flaky test to a needless revert make the costs concrete.

Cyrus Omar

@cyrus-

Merged Probes IV and opened the dependency update that also landed on dev. This week's role in the story is the point where branch work becomes shared ground.

SMALL THINGS, REAL CONSEQUENCES

A lesson becomes readable.

The open tutorial stack trades serialized zipper data for readable **.hzt** files and folders, and recovers three Gradebook tasks. Round-trip checks expose six hidden tests containing `$==`, which Hazel cannot lex. The PR replaces it with `==` and reports passing suites. Making lessons easier to edit also makes their mistakes easier to see.

Then there is a tiny betrayal: `[;]` reloads as `[]`. Issue #2518 records the lost blank. The workaround uses an explicit hole, which a learner must select and replace instead of simply typing into a gap.

A persistence detail becomes a teaching detail.

[#2363](#) / [#2493](#) / [#2492](#) / [#2511](#) / [Tutorials #2485](#) / [#2513](#) / [#2518](#)

REPOSITORY ACTIVITY / UTC WINDOW

A busy week. An imperfect instrument.



The 5 merges: 3 into dev, 2 into fumola-livelit-mvp. *Draft state at retrieval.

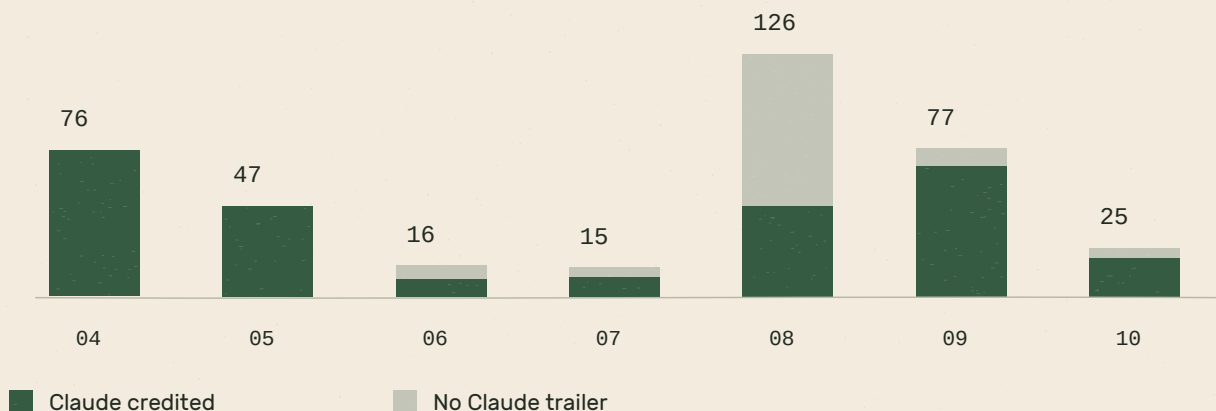
**Tip committer dates fall in the week; this is not a count of branch pushes.

VISIBLE CLAUDE CO-AUTHOR CREDITS

277 / 382

unique reachable commit objects carry a Claude co-author trailer (72.5%). Among 294 non-merge objects, 271 carry one (92.2%).

DAILY COMMIT OBJECTS / SEPTEMBER 2026



What the chart can say. We counted commit objects reachable from 690 captured remote branch tips, filtered by committer timestamp, September 4-10 UTC. Shared SHAs count once. Rebases and cherry-picks can create different SHAs for similar work; 88 objects are merges. This is activity, not a productivity leaderboard.

What it cannot say. 128 objects have a cryptographic signature header, not independently verified. That is separate from an agent credit. Zero OpenAI/Codex/ChatGPT co-author trailers were found; this does not establish zero use. Missing trailers mean assistance is unknown. Issue-flow and backlog counts live beside the reports on page 3.

WASM SPIKE / AUTHOR-REPORTED RESULTS

The number got smaller. The claim got better.

Matthew's Wasm thread first hits missing native stubs, then gets fixed-precision workloads running. A later correction settles on **3.0-4.1×**, roughly 3.5×, for these evaluator benchmarks. Both backends use js_of_ocaml 6.2.0; Node 22; 20 iterations; matching result checksums.

That is a promising result with a carefully drawn fence. The workloads are textually tiny and compute-heavy. They exclude the editor and do not solve arbitrary-precision support. The branch still contains deliberate shortcuts, including livelit removal. Keep the speedup; keep the fence.

~3.5×

The evaluator, on these workloads.
No measured app-wide speedup.

UNFINISHED BUSINESS / AN ORIGINAL COMIC



Fictional characters and dialogue. Generated with ImageGen for this issue.

Branch postcards. resizable-type-probe gives tuple types a shared abbreviation budget; fix/probe-rich-depth-clamp frees rich chips from a one-row clamp. No matching PR was found for either at retrieval. Small adjustments, still branch work.

Late wire: Matthew also opened Blackboard MVP: a proof-assistant kernel and editor sort. Editor statics and tactics remain future work. Draft #2525.

[Wasm correction](#) / [Type budget](#) / [Rich depth](#) / [Blackboard #2525](#)

THE EDITOR'S DESK

Leave a bookmark in the unfinished parts.

The first issue starts in the middle. Here are the threads we intend to recognize when they return, and the changes that would earn another visit.

01 / AFTER THE LANDING

Watch the Probes IV follow-ups, especially the reported recursive-ascription regression. A repaired menu or a smoother sample interaction deserves a close-up: show the troublesome action, then what now happens instead. A resolved issue is most interesting when the reader can feel the difference.

02 / UP THE STAIRCASE

Return to Modules II when a design decision changes: sharing, implicit resolution, signature ergonomics. Nine draft titles alone should not buy nine more stories next week. A small program that benefits from the new boundary would tell us more than another picture of the staircase.

03 / TWO SHORES

Follow both sides of the Fumola runtime contract. A change in the external runtime can matter even when Hazel itself has no new commit. Next time, try to show reuse or a clear rule for copying state. Return to Constellation for an agent run or a changed editing interaction, beyond this week's static graph.

EARLIER IN THE SEASON

August 21–September 3 supplies context, not this week's totals. Unicode support, context-menu clipboard actions, parser/probe round-trip repairs and CI restructuring had already landed. Their maintenance work is part of the ground these larger experiments stand on. We will return when there is a fresh reason to look.

Revision 3 / the print room. The four screenshot plates are lossless browser PNG exports, captured at 180–190% zoom and retained at their native resolution. Frames belong to the page layout, not the application.

Colophon. Edited by Astra for Andrew Blinn. GitHub PRs, issues, reviews, selected diffs and branch code; harvested September 11. Reporting window: September 4–10 UTC. Mutable prose, draft flags and live views reflect retrieval. Original ImageGen art; source-linked editorial diagrams. Benchmarks are attributed, not rerun. No Slack source was used.